



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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ASSIGNMENT No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.



Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

```
1 class BankAccount {
2     final int accountNumber;
3     String accountHolder;
4     double balance;
5     BankAccount(int accountNumber, String accountHolder, double balance) {
6         this.accountNumber = accountNumber;
7         this.accountHolder = accountHolder;
8         this.balance = balance;
9     }
10    void displayAccount() {
11        System.out.println("Account Number: " + accountNumber);
12        System.out.println("Account Holder: " + accountHolder);
13        System.out.println("Balance: " + balance);
14    }
15 }
16 class Main {
17     public static void main(String[] args) {
18         BankAccount account = new BankAccount(1001, "Karan", 5000000.0);
19         account.displayAccount();
20         // account.accountNumber = 2002;
21         // This will give an error because accountNumber is final.
22     }
23 }
24 }
```

Problems Javadoc Declaration Console Properties
:terminated> Main [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (13-Aug-2026, 11:35:38 pm)
Account Number: 1001
Account Holder: Karan
Balance: 5000000.0

```
Account Number: 1001
Account Holder: Karan
Balance: 5000000.0
```



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2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
1 class LibraryBook {
2     final String ISBN;
3     String title;
4     String author;
5     double price;
6     LibraryBook(String ISBN, String title, String author, double price) {
7         this.ISBN = ISBN;
8         this.title = title;
9         this.author = author;
10        this.price = price;
11    }
12    void displayBook() {
13        System.out.println("ISBN: " + ISBN);
14        System.out.println("Title: " + title);
15        System.out.println("Author: " + author);
16        System.out.println("Price: " + price);
17    }
18 }
19 class Main2 {
20     public static void main(String[] args) {
21         LibraryBook book = new LibraryBook(
22             "978-0135166307",
23             "Java Programming",
24             "Karan Dave",
25             650.0
26         );
27         book.displayBook();
28     }
29 }
30
```

Console Output:

```
<terminated> Main2 [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (13-Aug-2026, 11:40:30 pm)
```

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